Step1:

Write the code in C language file p1 inside project directory

```c
#include <linux/init.h>
#include <linux/module.h>
#include <linux/kernel.h>

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Atekah");
MODULE_DESCRIPTION("Kernel module to sort user input numbers in descending order");
MODULE_VERSION("0.3");

// Input parameters
static int input_numbers[5] = {0, 0, 0, 0, 0};
static int input_count = 0;
module_param_array(input_numbers, int, &input_count, S_IRUGO);
MODULE_PARM_DESC(input_numbers, "An array of integers provided by user");

static void sort_numbers_desc(int *array, int count)
{
    int i, j, temp;
    for (i = 0; i < count - 1; i++) {
        for (j = 0; j < count - i - 1; j++) {
            if (array[j] < array[j + 1]) {
                // Swap elements
                temp = array[j];
                array[j] = array[j + 1];
                array[j + 1] = temp;
            }
        }
```

```c
    }
}

static int __init numbers_module_init(void)
{
    int i;
    printk(KERN_INFO "Sorted Numbers Module Loaded.\n");

    printk(KERN_INFO "Received %d numbers from user.\n", input_count);

    // Sort numbers descending
    sort_numbers_desc(input_numbers, input_count);

    for (i = 0; i < input_count; i++) {
        printk(KERN_INFO "Sorted Number[%d]: %d\n", i, input_numbers[i]);
    }

    return 0;
}

static void __exit numbers_module_exit(void)
{
    printk(KERN_INFO "Sorted Numbers Module Unloaded.\n");
}

module_init(numbers_module_init);
module_exit(numbers_module_exit);
```

Note : file name is project.c

Step 2:

Making the Makefile

```
obj-m += p1.o

all:
    make -C /lib/modules/$(shell uname -r)/build M=$(PWD) modules

clean:
    make -C /lib/modules/$(shell uname -r)/build M=$(PWD) clean
```

Step 3:

Write make in terminal to produce p1.ko

```
ubuntu@ubuntu:~$ mkdir project
ubuntu@ubuntu:~$ cd project
ubuntu@ubuntu:~/project$ nano p1.c
ubuntu@ubuntu:~/project$ nano Makefile
ubuntu@ubuntu:~/project$ make
make -C /lib/modules/6.8.0-41-generic/build M=/home/ubuntu/project modules
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-41-generic'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.2.0-23ubuntu4) 13.2.0
  You are using:           gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0
  CC [M]  /home/ubuntu/project/p1.o
  MODPOST /home/ubuntu/project/Module.symvers
  CC [M]  /home/ubuntu/project/p1.mod.o
  LD [M]  /home/ubuntu/project/p1.ko
  BTF [M] /home/ubuntu/project/p1.ko
Skipping BTF generation for /home/ubuntu/project/p1.ko due to unavailability of vmlinux
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-41-generic'
ubuntu@ubuntu:~/project$ ls
Makefile        modules.order  p1.ko   p1.mod.c  p1.o
Module.symvers  p1.c           p1.mod  p1.mod.o
```

p1.ko

Step 4:

Write sudo insmod p1.ko in terminal with the parameters and sudo dmesg to display the message

```
ubuntu@ubuntu:~/project$ sudo insmod p1.ko input_numbers=12,30,42,1,22
```

```
ubuntu@ubuntu:~/project$ sudo dmesg | grep "Sorted"
```

```
[ 2485.977410] Sorted Numbers Module Loaded.
[ 2485.977423] Sorted Number[0]: 42
[ 2485.977426] Sorted Number[1]: 30
[ 2485.977429] Sorted Number[2]: 22
[ 2485.977432] Sorted Number[3]: 12
[ 2485.977435] Sorted Number[4]: 1
```

Step 5:

Write sudo rmmod project to remove it and sude dmesg to display another message

```
ubuntu@ubuntu:~/project$ sudo rmmod p1
```

```
ubuntu@ubuntu:~/project$ sudo dmesg | grep "Sorted"
[ 2485.977410] Sorted Numbers Module Loaded.
[ 2485.977423] Sorted Number[0]: 42
[ 2485.977426] Sorted Number[1]: 30
[ 2485.977429] Sorted Number[2]: 22
[ 2485.977432] Sorted Number[3]: 12
[ 2485.977435] Sorted Number[4]: 1
[ 2827.371184] Sorted Numbers Module Unloaded.
```

Conclusion

This Linux kernel module that accepts user input, processes the data, and outputs results to the kernel log.By using functions like printk.